Part A: Complete each question, showing work in the space provided. Do not attach additional pages. Each question in this part is worth 5 pts.

1. Use back-substitution to solve the system of linear equations. Write your solution as an ordered triple.

$$-x + y - z = 0$$
$$7y + z = 5$$
$$\frac{1}{7}z = 0$$

2. Explain why the given system of equations must have at least one solution (without solving the system).

$$3x + 3y - z = 0$$
$$6x + 3y + 6z = 0$$
$$3x + 9y - 17z = 0$$

Part B: Complete each question, showing work in the space provided. Do not attach additional pages. Each question in this part is worth 10 pts.

3. Use an inverse matrix to solve the system of linear equations. Write your solution as an ordered pair.

$$\begin{aligned} x_1 + 5x_2 &= -2 \\ -5x_1 - 4x_2 &= -11 \end{aligned}$$

4. Solve the system of equations using either Gaussian elimination with back-substitution or Gauss-Jordan elimination. If there are no solutions, write No Solution. If there are an infinite number of solutions, express x_1, x_2, and x_3 in terms of the parameter t.

$$\begin{aligned} x_1 + x_2 - 5x_3 &= 3 \\ x_1 \qquad - 2x_3 &= 1 \\ 2x_1 - x_2 - x_3 &= 0 \end{aligned}$$

5. Find the inverse of the matrix.

$$\begin{bmatrix} 5 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 5 \end{bmatrix}$$

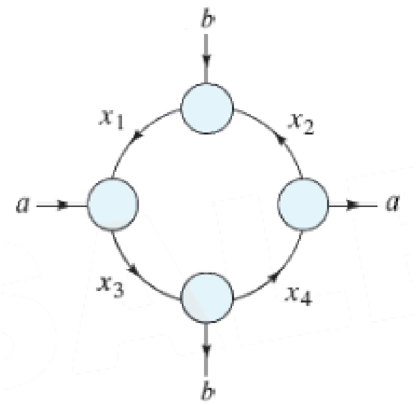
Part C: Application Questions (10 pts each). Choose 6 to solve. You may choose to solve an additional 1 question for up to 5 points of extra credit on this exam. Cross off the questions you don't want considered in your grade. Circle the question you want considered for extra credit.

For questions in part C, I do NOT need to see row operations. Show the equations when applicable, the initial matrix, and the row-reduced matrix.

1. Two planes start from Los Angeles International Airport and fly in opposite directions. The second plane starts $\frac{1}{2}$ hour after the first plane, but its speed is 20 kilometers per hour faster. Two hours after the first plane departs, the planes are 3110 kilometers apart. Find the airspeed of each plane.

2. Determine the polynomial of least degree whose graph passes through the points $(-1, 2), (0, 0), (1, 1), (4, 58)$.

3. The figure shows the flow of traffic (in vehicles per hour) through a network of street. Assume $a = 300$ and $b = 600$. Solve this system for the variables x_1, x_2, x_3, x_4. If the system has an infinite number of solutions, express the variables in terms of the parameter t. Then find the traffic flow when $x_1 = 2x_2$.



4. Use a system of equations to find the partial fraction decomposition of the rational expression. Then solve the system using matrices.

$$\frac{4x^2 + 3x - 3}{(x-1)(x+1)^2} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$$

5. The market research department at a manufacturing plant determines that 30% of the people who purchase the plant's product during any month will not purchase it the next month. On the other hand, 40% of the people who do not purchase the product during any month will purchase it the next month. In a population of 1000 people, 100 people purchased the product this month. How many will purchase the product next month and in 2 months?

6. In a population of 400,000 people, 120,000 are infected with a virus. After a person becomes infected and then recovers, the person is immune (cannot become infected again). Of the people who are infected, 4% will die each year and the others will recover. Of the people who have never been infected, 30% will become infected each year. How many people will be infected in 4 years? Show how you set up this problem, but feel free to use technology for the matrix multiplication. (Round your answer to the nearest whole number.)

7. A system composed of two industries, coal and steel, has the following input requirements.

   (a) To produce $1.00 worth of output, the coal industry requires $0.20 of its own product and $0.40 of steel.
   (b) To produce $1.00 worth of output, the steel industry requires $0.30 of its own product and $0.60 of coal.

   Find D, the input-output matrix for the system. Then solve for the output matrix X in the equation $x = DX + E$, where E is the external demand matrix $E = \begin{bmatrix} 10{,}000 \\ 20{,}000 \end{bmatrix}$. Round to the nearest whole number.

8. A fuel refiner wants to know the demand for a grade of gasoline as a function of price. The table shows daily sales y (in gallons) for three different prices. Find the least squares regression line for these data, then estimate the demand when the price is at $3.65/gallon.

Price, x (dollars per gallon)	$3.25	$3.50	$3.75
Demand, y (gallons)	4500	3750	3300